

Periodic Table of Elements vAsbestos

Super Seven

HI
HBr
HCl
HNO₃
H₂SO₄
HClO₃
HClO₄

Gas

NO
NO₂
CO
CO₂
CH₄
C₂H₆
C₃H₈
C₄H₁₀
N₂O
NH₃
SO₃
SO₂
H₂S
HCl

Conversions

1 L · atm = 101.3 J
K = °C + 273.15
°C = $\frac{5}{9} (°\text{F} - 32)$
°F = $\frac{9}{5} \text{°C} + 32$
1 cal = 4.184 J
1 lb = 453.59 g
1 atm = 760 mmHg
= 760 torr = 101.325 kPa
1 bar = 10⁵ Pa = 10⁵ N/m²

Periodic Trends

$\mathbb{Z}_{eff}$ increase →↓
EN, IE, & EA increase →↑
Radius & Metallic decrease →↑

Equilibrium

When $aA + bB \rightleftharpoons cC + dD$,
 $K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$
 $K_p = \frac{(P_C)^c(P_D)^d}{(P_A)^a(P_B)^b}$
 $K_a = \frac{[H^+][A^-]}{[HA]}$
 $K_b = \frac{[OH^-][HB^+]}{[B]}$
 $K_w = K_aK_b = [H^+][OH^-]$
 $K_w = 1.0 \times 10^{-14}$ (25° C)
pH = pK_a + log $\frac{[A^-]}{[HA]}$ = −log[H⁺]
pH + pOH = 14.
pK_a = −log K_a, pK_b = −log K_b.

Gasses/Solutions

$PV = nRT$
 $P_A = P_{\text{total}}X_A$, where $X_A = \frac{\text{moles } A}{\text{total moles}}$
 $P_{\text{total}} = P_A + P_B + P_C + \dots$
 $M = \frac{\text{moles solute}}{\text{Liters solution}}$, $m = \frac{\text{moles solute}}{\text{kg solvent}}$
 $M_1V_1 = M_2V_2$ for dilution
STP = 273.15 K and 1.0 atm
At STP, ideal gas 22.4L mol^{−1}.
Standard conditions 25° C, 1 atm.
 $v_{rms} = \sqrt{\frac{3RT}{\mathcal{M}}}$

Quantum

$E_{\text{photon}} = hf = \frac{hc}{\lambda} \implies c = \lambda f$
 $\lambda = \frac{h}{mv}$
 $R_{H_{\text{Rydberg}}} = 1.097 \times 10^7 \text{ m}^{-1}$
 $\frac{1}{\lambda} = R_H \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$
 $\Delta E = (-2.18 \times 10^{-18} \text{ J}) \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$

Thermo/Electrochem

$q = mc\Delta T$, $\Delta E = q + w$, $H = E + PV$
 $\Delta S^\circ = \sum_{\text{products}} S^\circ - \sum_{\text{reactants}} S^\circ$
↑ Likewise for ΔH° and ΔG° ↑
 $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$
 $= -RT \ln K = -nFE^\circ$
 $I = \frac{q}{t}$
 $E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{RT}{nF} \ln Q$.

Kinetics

$[A]_t - [A]_0 = -kt$ (0th order)
 $\ln[A]_t - \ln[A]_0 = -kt$ (1st order)
 $\frac{1}{[A]_t} - \frac{1}{[A]_0} = kt$ (2nd order)
 $t_{1/2} = \frac{0.693}{k}$ (1st order)

Solutions (cont.)

$\Delta T_b = K_bmi$ $\Delta T_f = -K_fm_i$
 $P_A = X_AP_A^\circ$ $S_g = kP_g$
 $\Pi = \left(\frac{n}{V} \right) RT = MRT$
 $(K_{b_{\text{water}}}, K_{f_{\text{water}}}) = (0.512, 1.86)^\circ\text{C/m}$
 $\Delta H_{\text{fus water}} = 6.008 \text{ kJ/mol}$
 $\Delta H_{\text{vap water, 100}^\circ\text{C}} = 40.67 \text{ kJ/mol}$
 $c_{\text{ice}} = 2.093 \text{ J/(}^\circ\text{C)}$
 $c_{\text{water}} = 4.184 \text{ J/(}^\circ\text{C)}$
 $c_{\text{steam}} = 1.841 \text{ J/(}^\circ\text{C)}$

18 VIIIA

2
He
Helium
4.00

13 IIIA

5 2.04
B
Boron
10.81

14 IVA

6 2.55
C
Carbon
12.01

15 VA

7 3.14
N₂
Nitrogen
14.01

16 VIA

8 3.44
O₂
Oxygen
16.00

17 VIIA

9 3.98
F₂
Fluorine
19.00

18 VIIIA

10
Ne
Neon
20.18

3 IIIB

11 0.98
Li
Lithium
6.94

4 IIA

4 1.57
Be
Beryllium
9.01

11 IA

3
Na
Sodium
22.99

12 IIA

12 1.31
Mg
Magnesium
24.31

19 IA

19
K
Potassium
39.10

20 IIA

20 1.00
Ca
Calcium
40.08

21 IIIB

21 1.36
Sc
Scandium
44.96

22 IVB

22 1.54
Ti
Titanium
47.87

23 VB

23 1.63
V
Vanadium
50.94

24 VIB

24 1.66
Cr★
Chromium
52.00

25 VIIB

25 1.55
Mn
Manganese
54.94

26 VIIIB

26 1.83
Fe
Iron
55.85

27 VIIIB

27 1.88
Co
Cobalt
58.93

28 VIIIB

28 1.91
Ni
Nickel
58.69

29 VIIIB

29 1.90
Cu★
Copper
63.55

30 VIIIB

30 1.65
Zn
Zinc⁽²⁺⁾
65.38

31 VIIIB

31 1.81
Ga
Gallium
69.72

32 VIIIB

32 2.01
Ge
Germanium
72.63

33 VIIIB

33 2.18
As
Arsenic
74.92

34 VIIIB

34 2.55
Se
Selenium
78.97

35 VIIIB

35 2.96
Br₂
Bromine
79.90

36 VIIIB

36 3.00
Kr
Krypton
83.80

37 IA

37 0.82
Rb
Rubidium
85.47

38 IIA

38 0.95
Sr
Strontium
87.62

39 IIIB

39 1.22
Y
Yttrium
88.91

40 IVB

40 1.33
Zr
Zirconium
91.22

41 VB

41 1.6
Nb★
Niobium
92.91

42 VIB

42 2.16
Mo★
Molybdenum
95.95

43 VIIB

43 1.9
Tc
Technetium
(98)

44 VIIIB

44 2.2
Ru★
Ruthenium
101.07

45 VIIIB

45 2.28
Rh★
Rhodium
102.91

46 VIIIB

46 2.20
Pd★★
Palladium
106.42

47 VIIIB

47 1.93
Ag★
Silver⁽¹⁺⁾
107.87

48 VIIIB

48 1.69
Cd
Cadmium
112.41

49 VIIIB

49 1.78
In
Indium
114.82

50 VIIIB

50 1.96
Sn
Tin
118.71

51 VIIIB

51 2.05
Sb
Antimony
121.76

52 VIIIB

52 2.1
Te
Tellurium
127.60

53 VIIIB

53 2.86
I₂
Iodine
126.90

54 VIIIB

54 2.60
Xe
Xenon
131.29

55 IA

55 0.79
Cs
Caesium
13